Mobi²Still: People Detection and Tracking With Mobile Human-Equipped mmWave Radars

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***Abstract*—Due to the ability to penetrate darkness, smoke, and fog, wireless sensing technologies offer unique advantages for real-world deployments. Recent studies have mounted wireless sensing devices on mobile systems (such as drones and wheeled robots). Despite their promising potential, these approaches are predominantly designed for non-human platforms that exhibit limited mobility, typically involving translational movement with minimal rotation. When deployed on human carriers, however, the frequent body rotations and complex motion patterns significantly degrade their performance, due to the unpredictable changes in target position. In this paper, we propose Mobi²Still, which utilizes mobile human-equipped millimeter wave (mmWave) radars for people detection and tracking. The core idea is to match recurring environment information across different times to estimate radar motion in reverse, and based on this, enable accurate people detection and tracking. Specifically, Mobi²Still extracts geometric structure of scene objects which is radar-motion-independent to match the recurring environmental objects, thereby adapting both rotation and translation of mmWave radars. Furthermore, to improve Mobi²Still's generalization, we design a velocity-aware calibration mechanism to guide it to focus on scene-independent motion features. Experiments demonstrate that Mobi²Still can accurately detect and track people with 98.10% F1-score and 98.32% MOTA, achieving at least 7.54% F1-score (and 11.20% MOTA) improvements compared with state-of-the-art approaches.**

***Index Terms*—Wireless sensing, mobile mmWave radar, people detection and tracking, device motion.**

I. Introduction

PEOPLE detection and tracking applications have become prevalent in recent years [1], [2], ranging from crowd analytics [3], [4] to security [5], [6]. Among these, *mobile sensing using human-equipped devices* serves as a key component. For instance, a human-carried system that senses the positions of nearby pedestrians allows visually impaired individuals to navigate safely [7]; Tracking victims in smoke-filled fire zones can greatly enhance firefighters' search efficiency and reduce casualties. Moreover, although vision-based [8], [9] and LiDAR-based [10], [11], [12] approaches have shown promise in common scenes, wireless sensing offers unique advantages in penetrating darkness, dense smoke, rain, and fog, highlighting its immense potential in practice.

Previous wireless sensing works rely on stationary devices [1], [13], [14], [15], [16]. Recently, researchers have begun exploring wireless sensing with mobile platforms, including vehicles [17], [18], [19], drones [20], wheeled robots [21], [22], or be held in human hands [23], [24], [25]. In these studies, wireless sensors (such as LoRa receivers and millimeter wave radars) are mounted on the platforms, where their movement patterns are synchronized with the platform's motion. Although recent works perform well on their dedicated platforms, where sensors' movement is primarily translational with rare rotation, *their performance degrades significantly when deployed on human body*. Specifically, when a human walks, the entire body undergoes *translational motion and frequent rotation* [26]. Existing methods typically depend on tracking each target's continuous motion, where tracking is based on predicting each target's next positions and associating the predicted target positions with actual detected objects across different times [27], [28], [29]. However, frequent rotation of human-equipped radars leads to drastic changes in target positions, significantly increasing the difficulty of predicting the future locations. This severely misleads detection associations and ultimately causes the failure of target tracking.

To address the above issue, we propose Mobi²Still, a robust people detection and tracking system based on human-equipped moving millimeter-wave (mmWave) radars. At a high level, instead of tracking based on unpredictable positions used in traditional approaches [17], [18], [20], [21], [22], [23], [24], [25], *Mobi²Still uses the detected objects' geometric structures to eliminate the impact of frequent rotation*. The geometric structures sketch the spatial relationships between detected objects in the scene, e.g., relative distance, direction, and angle, which maintain constant and still during radar rotation. In other words, radar rotation does not affect the intrinsic spatial relationships between objects. Based on above findings, we elaborate Mobi²Still's core idea and usage scenario in Fig. 1. Specifically, the human-equipped mmWave radar first perceives the surrounding environment and the moving person in the form of point clouds, which characterize their positions and velocities relative to the radar. Then, leveraging the geometric structures, Mobi²Still matches the same (i.e., recurring) environmental

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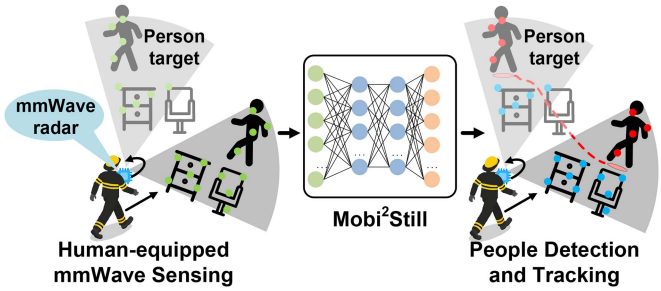
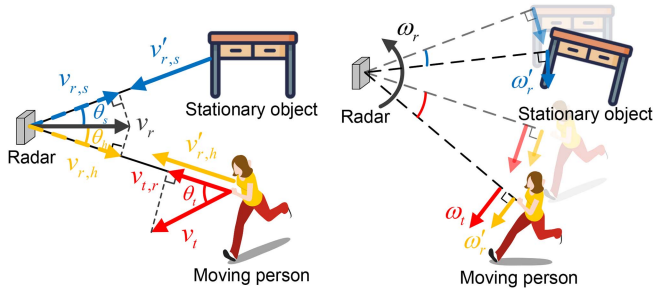


Fig. 1. Usage scenario: A walking human is equipped with a mmWave radar in both the translational and rotational states. By employing Mobi²Still, the person target is accurately detected and tracked from the environment.



(a) Radar translation motion. (b) Radar rotation motion.

Fig. 2. The sensing model for radar motion, which can be decomposed into translation and rotation [33].

objects at any two different time instants to estimate the radar's motion. Finally, Mobi²Still reversely detects and tracks people from the environment based on the radar motion. To realize Mobi²Still, we identify and address two main challenges as follows.

How to match hard-to-find recurring environments across different time? When equipped on humans, the mmWave radar undergoes frequent rotations, which causes substantial changes in the sensed scene information, making it difficult to match the same environments recurring at any two time instants. This phenomenon arises from two factors: *(i)* Due to the rotation and specular reflections [30], the recurring environments are scarce; *(ii)* The rotation of the radar results in large positional distribution and orientational shifts of the recurring environmental portions at the radar's view. Therefore, we design a *geometric structure based recurring environment matching method*, consisting of a geometric structure extraction network module and an environment geometry match module. The first module extracts the geometric structures of the sensing scene, which depicts the relative spatial relationships utilizing relative distances and angles among all scene objects. Therefore, even when radar rotation causes significant changes in the position distributions and orientations, the extracted geometric structures still remain stable. The second module enables matching recurring environment based on geometric feature similarity even when the recurring portions are scarce, as the geometric structure of an environmental portion is unique, with distinct geometric features across different environmental portions.

How to enhance the generalization capability of Mobi²Still in unknown scenes? We observe that the people detection model based on recurring environment matching, which is trained in limited scenarios, tends to focus on scene-specific information, such as the environment layout and individuals encountered during training. This significantly affects Mobi²Still's application in unfamiliar scenarios. To overcome this challenge, we design *a velocity-aware calibration mechanism*, which enforces the model to focus on scene-independent velocity information. This mechanism learns the velocity correspondence between the radar motion and any static environment motions relative to the radar, and treats the points deviating from the relationship as people. By doing so, Mobi²Still can improve the model's generalization, especially in unseen environments and for unknown individuals.

We prototype Mobi²Still on a commercial off-the-shelf (COTS) Texas Instruments IWR6843ISK mmWave radar [31], while using an RGB-Depth camera (Femto Bolt [32]) as the ground truth. Overall, Mobi²Still achieves compelling performance when the radar is equipped on human under three real-world scenarios including the classroom, library, and office building. Results show the 98.36% accuracy and 98.10% F1-score for people detection, and 98.32% Multiple Object Tracking Accuracy (MOTA) for people tracking. Compared with state-of-the-art approaches [18], [19], [20], Mobi²Still improves at least 7.54% in F1-score and 11.20% in MOTA. Mobi²Still also achieves a good generalization with the F1-score dropping by only 1.42% upon unseen environments and 2.40% upon unknown individuals. Moreover, we conduct a real-world case study to illustrate Mobi²Still's unique advantages in penetrating dense smoke under an extreme scenario.

Contributions of Mobi²Still can be summarized as follows.

- We propose a novel human-equipped mobile mmWave sensing method for people detection and tracking. By matching recurring environments across different times based on geometric structure, it achieves effectiveness when the radar simultaneously translates and rotates.
- We design a velocity-aware calibration mechanism that relies on scene-independent velocity cues rather than scene-specific features, yielding cross-scene generalization to unseen environments and unknown individuals.
- We implement and evaluate Mobi²Still using a COTS mmWave radar. Extensive experiments demonstrate that Mobi²Still achieves excellent performance for people detection and tracking with human-equipped radars and shows impressive generalization capability.

II. Understanding People Detection With Human-Equipped Mobile Mmwave Radar

A. Sensing Model for Mmwave Radar Motion

The mmWave radar when moving undergoes complex rigid-body motions, which, according to Chasles theorem [33], can be decomposed into translational and rotational components. We next introduce radar sensing models for them.

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Translational model: As shown in Fig. 2(a), the radar translates at a velocity of v_r along a specific direction while transmitting signals to both a static environmental object and a person, which enables the radar to perceive their radial velocities and angles relative to itself. The angles between the radar's velocity direction and the signal transmission paths are denoted as θ_s and θ_h, respectively. The human moves with a translation velocity of v_t in a certain direction, with an angle θ_t between the velocity direction and related signal transmission path. The radial velocities of the static object v_{object} and the human v_{human} relative to the radar can be obtained by:

$$v_{\text{object}} = v'_{r,s} = -v_{r,s} = -v_r \cdot \cos\theta_s, \tag{1}$$

$$v_{\text{human}} = v_{t,r} + v'_{r,h} = v_{t,r} - v_{r,h} = v_t \cdot \cos\theta_t - v_r \cdot \cos\theta_h. \tag{2}$$

Rotational model: As illustrated in Fig. 2(b), the radar undergoes rotation with an angular velocity of ω_r. The human rotates around the radar, with an angular velocity given by $\omega_t = v_t \cdot \sin\theta_t$. The angular velocities of the static object ω_{object} and the human ω_{human} relative to the radar can be derived by:

$$w_{\text{object}} = w'_r = -w_r, \tag{3}$$

$$w_{\text{human}} = w_t + w'_r = w_t - w_r. \tag{4}$$

B. People Detection Using Human-Equipped Mmwave Radar

When equipped on a human, the motion of mmWave radar can be decomposed into continuous translation and rotation. From (1) to (4), the motion of static objects relative to the radar is solely influenced by the radar's own movement, whereas the motion of a person relative to the radar is affected by not only the radar but also the person's own motion. Thus, if the radar's movement can be determined, we can distinguish the static environment and people based on this movement.

Basic workflow: From above analysis, we drive Mobi^{2}Still as the following operations. *Firstly*, it matches the environmental portions that recur at any two time instants to obtain the whole environmental motion and deduce the radar's motion. *Secondly*, it differentiates between the static environment and people based on the derived radar motion. During the workflow, we infer the radar motion from radar data rather than using an inertial measurement unit (IMU). The reasons lie in that: *(i)* Prolonged usage can cause the IMU to loosen, leading to a shift in its relative position to the radar and false radar motion estimation; *(ii)* The IMU inevitably accumulates drift, increasing the error during long-term usage [34].

C. Challenges

While realizing Mobi^{2}Still, we meet two challenges.

1) C1 - Matching the Hard-to-Find Environmental Portions Recurring at Any Two Different Time Instants: Due to the frequent rotation of the human-equipped radar, the scene information perceived by the radar changes obviously over time, making it difficult to match the same environmental portions that recur at any two distinct time instants. This challenge manifests in two aspects.



(a) Scene information at an earlier time. (b) Scene information at a later time.

Fig. 3. An example: Due to the radar's rotation and specular reflection, there are few recurring environmental portions across different time instants.

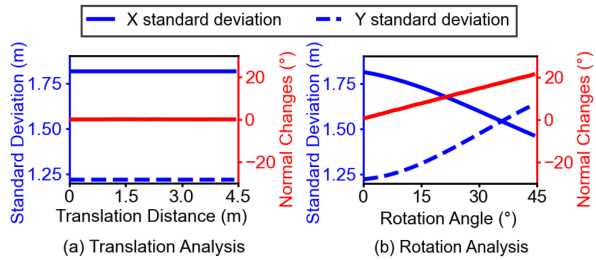


Fig. 4. Position distribution and orientation change of the same environmental point cloud under translational and rotational motion of radar.

Recurring environments are scarce: Due to the rotation of the radar and the effects of specular reflection [30], there are few environmental regions across different times. Fig. 3(a) and (b) respectively illustrate the scene's corresponding point cloud perceived by the radar at two distinct time instants when the radar rotates to the left, and the time of Fig. 3(b) follows that of Fig. 3(a). The dots represent the point cloud.

We can observe that: *(i)* As the radar rotates, the lost area leaves the radar's field of view while the new area enters, resulting in a gradual decrease in the recurring environmental portions as the rotation angle increases; *(ii)* The emitted mmWave signals undergo specular reflection on the surfaces of objects, and most of them fail to return to the radar, leading to missing portions in Fig. 3(b). Consequently, even if certain objects are within the radar's field of view, specular reflections may prevent them from being captured, further reducing the amount of recurring environmental portions.

Recurring environments change significantly in spatial features: The rotation of the radar causes obvious changes in spatial features of the recurring environmental portions in the radar coordinate system, making them difficult to match. Specifically, rotation causes larger displacement differences for more distant points, which disrupts the consistency of spatial features (in terms of position distribution and orientation) and increases the likelihood of mismatches. In contrast, translation leaves the point cloud structure unchanged, preserving feature consistency and resulting in accurate matching.

Fig. 4(a) and (b) analyze the position distribution and orientation changes of the environmental point cloud under translational and rotational motion, respectively. To quantify the position distribution, we calculate the standard deviations of the point cloud along the x and y axes. The normals indicate the orientation

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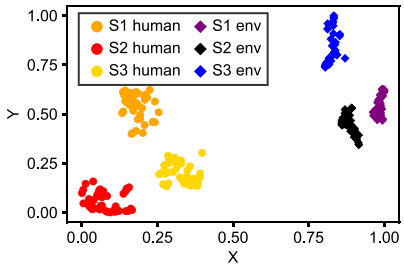


Fig. 5. The visualization of environmental and human features.

of each point, which are computed using Principal Component Analysis (PCA) [35]. We record the angular changes of these normals under different radar motions to capture the changes in orientation.

We can observe that: *(i)* As shown in Fig. 4(a), during translation, both the position distribution and orientation do not change, and simple translational alignment can effectively match the recurring portions; *(ii)* However, as depicted in Fig. 4(b), during rotation, both the standard deviations and the point cloud normal angles change significantly with increasing rotation angle, rendering the matching of recurring environmental portions challenging.

2) C2 - Lack of Generalization Across Unseen Scenes: After matching recurring environment to estimate radar motion, the system accurately detects people. However, the model exhibits limited generalization in unseen scenes, as training on data from a limited set of scenes tends to cause the model to focus on scene-related information, such as environmental layouts and personnel encountered during training. When exposed to new scenes with unfamiliar environments and humans, the model performs poorly, indicating its weak generalization.

To understand this challenge, we conduct an observation by visualizing the environmental and human features from different scenes with t-Distributed Stochastic Neighbor Embedding (t-SNE) [36], as shown in Fig. 5. Environmental and human features are largely separated, located in the upper right and lower left regions respectively, which indicates that the model can distinguish between the two. But the environmental and human features from the same scene are respectively highly clustered due to the inclusion of scene-specific information. This clustering indicates that the model overfits the training scenes. It mistakenly uses scene-related cues, rather than the intrinsic, scene-independent differences between humans and static environments, as the basis for discrimination. As a result, when the model encounters a new scene, it attempts to rely on scene-specific information, leading to poor performance.

III. System Overview

Mobi²Still is focused on accurate and generalized people detection and tracking under the disturbance of the environment with human-equipped mobile mmWave radars. Fig. 6 shows its system overview, consisting of data acquisition, people detection, and people tracking. Mobi²Still captures all scene objects including people and environment from mmWave radar in the form of point clouds, and detects people from the environment based on radar motion, which is estimated by matching the recurring environment across different times. Then, Mobi²Still keeps tracking people based on the detected human points.

For the core design, we propose to match hard-to-find recurring environment based on geometric structure. Thus, Mobi²Still can utilize the matched environment for accurate people detection. However, the model has insufficient generalization in unseen scenarios. To solve this issue, we design a velocity-aware calibration mechanism, which guides the model to focus on scenario-independent velocity. We now proceed to introduce the designs as follows.

Recurring Environment Matching: We design a geometric structure extraction module to extract the scene geometry, and an environment geometry match module to match the recurring environment based on the geometry similarity. Leveraging the extracted geometric structure, the recurring environment can be matched even when radar undergoes continuous translation and rotation. To match the same environmental portions across different times, Mobi²Still employs the following components: *(i)* Point geometric structuring, which is an encoder that extracts the local geometric structure for each point, yielding a local perspective. *(ii)* Scene geometric construction integrates the local geometric features of all points to construct the overall geometric structure of the scene. *(iii)* Environment geometry match uses a self-attention mechanism to identify similar geometric structures between scenes at different time instants, thereby matching the same environmental portions.

Velocity-aware Calibration: We design a velocity-aware calibration mechanism to enhance the model's generalization. To guide the model to distinguish between environment and people utilizing scene-independent velocity cues, Mobi²Still employs the following components. *(i)* Radar motion and environment correspondence: This component forces the model to establish a correspondence in the velocity dimension between the radar motion and the relative motion of static environmental objects, thereby preventing the extraction of scene-specific features. *(ii)* Human-environment classifier: It examines whether the velocity of any given point conforms to the established correspondence, classifying points that do not match as human.

IV. System Design

A. Geometric Structure Based Recurring Environment Match

To tackle the challenge of matching hard-to-find recurring environmental components that appear at any two different time instants (discussed in Section II-C1), we focus on extracting a kind of feature for each environmental component that is both stable under the radar's complex motion and sufficiently distinctive. Ultimately, we choose to extract the feature of geometric structure. This choice is motivated by the intuitive idea that, although the position distribution and orientation of an environmental component changes dramatically in the radar's view, its intrinsic geometric structure remains invariant. Furthermore, since different environmental components exhibit distinct geometric features, even a limited set of recurring environmental

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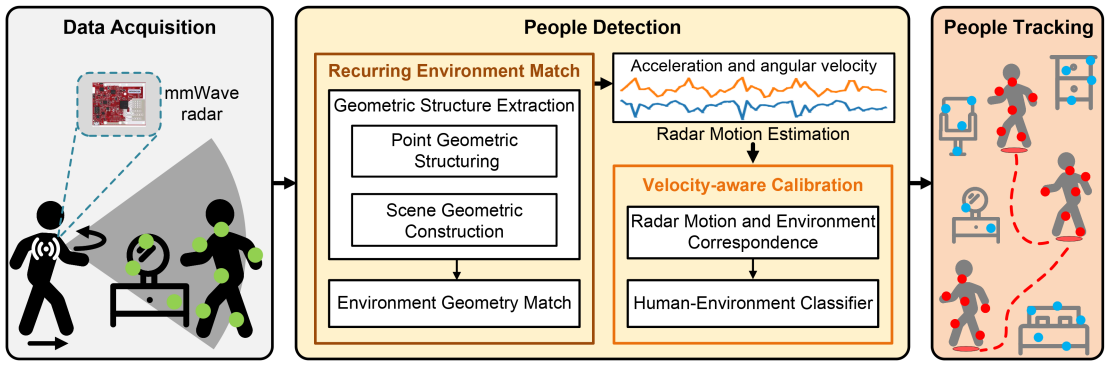


Fig. 6. The overview of Mobi^{2}Still.

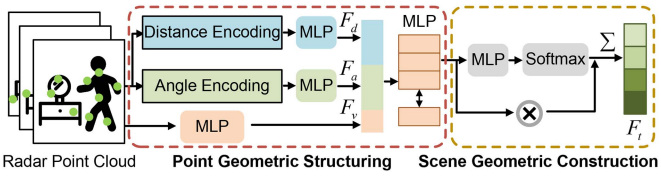


Fig. 7. Model architecture for geometric structure extraction.

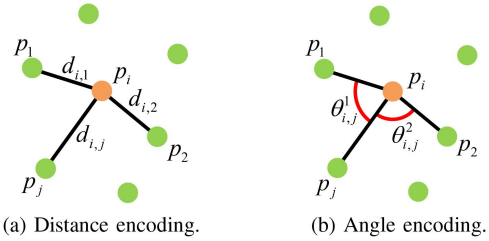


Fig. 8. The illustration for distance encoding and angle encoding.

elements can be successfully matched based on their geometric similarity.

In this section, we extract the geometric structural features of the scene and match the recurring environment based on the geometry features to estimate radar motion. The details of each component are as follows.

1) Geometric Structure Extraction: To achieve accurate matching, we extract geometric features of the scene that remain invariant under radar motion. Specifically, we first extract local geometric features for each point, and then aggregate all points to construct the overall geometric structure of the scene. The detailed procedure is illustrated in Fig. 7.

Point Geometric Structuring: Mobi^{2}Still extracts the local geometric features of individual points. To capture geometric characteristics that remain invariant under radar motion, we design a novel encoding method for radar point clouds. The key idea is to employ distance encoding and angle encoding to represent the distances and angles between points, respectively. Fig. 8 illustrates the specific encoding methods for point cloud distances and angles.

We next consider an arbitrary point p_i in the point cloud, with 3D coordinates (x_i, y_i, z_i) in the radar coordinate system. As shown in Fig. 8(a), for another arbitrary point p_j with coordinates (x_j, y_j, z_j), the distance between these two points is computed as $d_{i,j} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2}$. As illustrated in Fig. 8(b), for another arbitrary point p_j with coordinates (x_j, y_j, z_j), the angle between p_i and p_j is represented by a set of angles $a_{i,j} = \{\theta^k_{i,j}\}^N_{k=1}$, where we select the N points p_k that are closest to p_i (with 3D coordinates (x_k, y_k, z_k)). Using the cosine laws, the angle formed by the three points is defined as:

$$\theta^k_{i,j} = \arccos\left(\frac{(p_j - p_i)\cdot(p_k - p_i)}{\|p_j - p_i\|\cdot\|p_k - p_i\|}\right). \quad (5)$$

This encoding of local geometric structure provides a stable and distinctive representation for each point, facilitating robust and accurate matching of recurring environmental regions.

After obtaining the distance representation via distance encoding, the model applies multilayer perceptron (MLP) layers [37] to obtain distance feature F_d. Specifically, the MLP maps the data into a higher-dimensional space, allowing the model to capture more complex distance-related geometric features. The output is denoted as $F_d = MLP(e^d; \theta_d)$, where e^d represents radar points after distance encoding and θ_d is the learnable parameters of the MLP.

After angle encoding, the model applies MLP layers to obtain the angle feature F_a. Unlike distance encoding, which encodes one representation for every pair of points, angle encoding generates N representations for each pair, where N corresponds to the number of the nearest neighbors of one point. Consequently, the model first performs a weighted summation of the multiple neighborhood angles with MLP, thereby focusing on the inter-point angular information. The subsequent operation is identical to the MLP layers after distance encoding. The output is denoted as $F_a = MLP(MLP(e^a; \theta_{a,1}); \theta_{a,2})$, where e^a represents radar points after angle encoding, $\theta_{a,1}$ and $\theta_{a,2}$ are the corresponding learnable parameters of the two MLPs. It is noteworthy that both the distance feature and the angle feature are critical for describing the geometric structure. As discussed in Section VI-D, the absence of either feature will lead to model performance degradation.

Although the model extracts local distance and angle features of points, these static geometric features cannot reflect the motion of the objects and thus are insufficient for estimating radar

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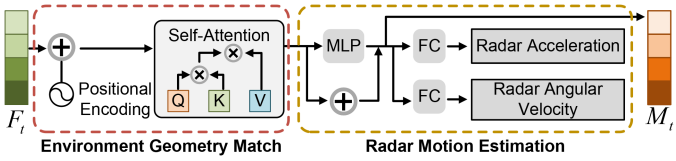


Fig. 9. Arch. of environment geometry match and radar motion estimation.

motion. Furthermore, the above manually designed geometric feature encoding and extraction may be constrained by predefined rules, resulting in the loss of some task-relevant geometric information. To address these issues, we employ an independent branch to process the raw point cloud data, including range, azimuth, elevation, and velocity, enabling the model to learn the motion trends of the scene while automatically extracting geometric features that manual design may overlook. From this branch, the model obtains feature $F_v = MLP(e^p; \theta_v)$, where e^p represents radar points and θ_v is the learnable parameters of the MLP.

To obtain richer geometric features, the model concatenates the features from the above three branches, and employs an MLP to transform the concatenated features into a higher-dimensional latent space by nonlinear operation, enabling more comprehensive geometric feature extraction. The output is a local geometric representation of each point denoted as $F_p = MLP([F_d \| F_a \| F_v]; \theta_p)$, where $\|$ represents concatenation and θ_p is the learnable parameters of the MLP.

Scene Geometric Construction: After obtaining the local geometric features for all points, this component is responsible for aggregating these local features into an overall geometric representation for subsequent environment geometry match. Given the local geometric features of the t-th frame as $P_t = \{p_{ij}^t \mid i \in [1, M], j \in [1, C]\}$ with feature dimensions (M, C), where M is the number of points and C denotes the dimensionality of the geometric features, and the overall scene geometric feature is defined as a C-dimensional vector F_t.

To adaptively shift the model's focus according to the input data, we employ an attention mechanism for aggregation. As illustrated on the right side of Fig. 7, we first use an MLP to reduce the feature dimension of P_t from C to 1, and then apply a softmax function to compute a weight n_i^t for each point. The weights for all points of the t-th frame are denoted as $N_t = \{n_i^t \mid i \in [1, M]\}$, which is formulated as $N_t = Softmax(MLP(P_t; \theta_n))$, where θ_n is the learnable parameters of the MLP. Using these weights, the overall scene geometric feature F_t is computed as the weighted sum of the local features as $F_t = \sum_{i=1}^{M} n_i^t \, p_i^t$, where $p_i^t = \{p_{ij}^t \mid j \in [1, C]\}$ represents the original geometric feature of each point. In this way, the MLP serves to assess the importance of each point in contributing to the overall scene geometry based on its local features, and the weights n_i^t, i.e., the model's focus can vary according to each point's geometric characteristics.

2) Environment Geometry Match: After obtaining the scene geometric structures F_t for all frames, Mobi²Still matches recurring environment segments that appear across different frames. As shown in Fig. 9, positional encoding [38] embeds temporal information into the features to prevent temporal confusion. This is crucial for matching the recurring segment across time, as the ordered sequence directly affects the motion trajectory of objects. Subsequently, we employ a multi-head self-attention mechanism [38] to capture cross-temporal dependencies from multiple perspectives. This allows the model to attend to features from all other frames, thereby identifying similar or identical geometry features for cross-temporal recurring environment matching. When radar undergoes translation and rotation, the combination of positional encoding and self-attention allows the model to consider both geometry similarity and temporal association, ensuring precise matching of the recurring environment.

Besides the static environment, the moving person is also present in the scene, so the scene geometric features also include the geometric structure of the moving human. As discussed in Sections II-A and II-B, the motion of recurring environmental components exhibits global regularity over time, resulting in stronger cross-temporal correlation, whereas human motion is influenced by individual movement himself or herself, thus more localized, leading to weaker cross-temporal correlation. By leveraging self-attention, the model successfully identifies the recurring environment with a stronger correlation and disregards the human and non-recurring environmental components with a weaker correlation. In addition, using cross-temporal correlation based on geometry similarity can avoid failures of RANSAC-LSQ [18], which occur when the number of human points with the same velocity exceeds that of static environmental points.

3) Radar Motion Estimation: After matching the recurring environment, Mobi²Still estimates the overall environmental motion and subsequently derives the radar motion. Since the environmental motion is merely the inverse of the radar's motion, the model directly estimates the radar motion. As shown in Fig. 9, the model first employs a residual connection [39] to preserve multi-level feature information and establish the relationship between environment matching and radar motion, resulting in the radar motion feature $M_t = MLP(\tilde{M}_t; \theta_r) + \tilde{M}_t$, where $\tilde{M}_t$ is the output representation of environment geometry match component, and θ_r is the learnable parameters of the MLP. Next, fully connected layers are used to obtain the radar's acceleration and angular velocity, which represent its translational and rotational motion, respectively.

It is important to note that acceleration and angular velocity have different dimensional units and numerical ranges. Combining them into a single task could lead to imbalanced gradient updates due to their scale differences, thereby affecting model optimization [40]. Therefore, we treat acceleration and angular velocity estimation as two separate tasks.

Loss of Radar Motion Estimation: The radar motion estimation is divided into two sub-tasks: radar acceleration and angular velocity estimation. The objectives of them are to minimize the error between the predicted radar acceleration and angular velocity and their corresponding ground truth, respectively. We minimize the two Mean Squared Error (MSE) losses:

$$L_A = \|\hat{a} - a\|^2, \tag{6}$$

$$L_W = \|\hat{w} - w\|^2, \tag{7}$$

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where $\|\cdot\|$ denotes L2-norm, $\hat{a}$ and a is the predicted acceleration and corresponding ground truth, $\hat{w}$ and w is the predicted angular velocity and ground truth.

B. Velocity-Aware Calibration

Based on the radar motion, the model can accurately detect people amid environmental interference. However, the model's focus on scene-specific features results in poor generalization in unseen scenes (discussed in Section II-C2). To address this challenge, we concentrate on uncovering the intrinsic, scene-independent differences between the environment and the human. We analyze the relationship between the radar's motion and the relative motion of static environmental objects and the moving person (discussed in Section II-A). From (1) to (4), we observe that there is a correspondence in the velocity dimension between the radar motion and the motion of static environmental objects, whereas no such correspondence exists for a person. Consequently, we select the velocity feature as the discrimination basis.

It is worth noting that doppler-based moving object detection approaches [41], [42] rely solely on the relative velocity relationship between the object and the radar. Therefore, these methods can only distinguish objects with different velocities rather than robustly estimate their actual velocity in the world coordinate system [43], [44]. In contrast, by accurately estimating the radar's own motion and leveraging its theoretical velocity correspondence with the static environmental objects, Mobi^{2}Still can determine the true velocity of each object, enabling more reliable detection.

Reason for relying on radial velocity rather than angular velocity: Due to the inherent sensing characteristics of the mmWave radar, it can only measure the radial velocity of a target relative to itself, but not the tangential velocity, i.e., the angular velocity. Estimating a target's angular velocity requires tracking the target and then indirectly calculating its angular velocity based on changes in its position and radial velocity. However, accurate target tracking is extremely challenging in our scenario, where the radar undergoes frequent rotations, thereby preventing the acquisition of the angular velocity.

1) Radar Motion and Environment Correspondence: After obtaining the radar motion feature and point cloud data, Mobi^{2}Still employs this component to establish a correspondence in the velocity domain between the radar motion and the relative motion of the static environment. In other words, based on the radar motion, the correspondence component enables the model to infer the radial velocity of a point belonging to the static environment at a specific position.

As illustrated in the upper part of Fig. 10, the model first uses an MLP to expand the dimensions of point cloud's positional information (range, azimuth, and elevation) to capture more complex positional features. The output is denoted as $F_{pos} = MLP(e^{pos}; \theta_{pos})$, where e^{pos} represents radar points' positions and θ_{pos} is the learnable parameters of the MLP. Next, the radar motion feature M_t is projected into the semantic space of point cloud and concatenated with each point's positional

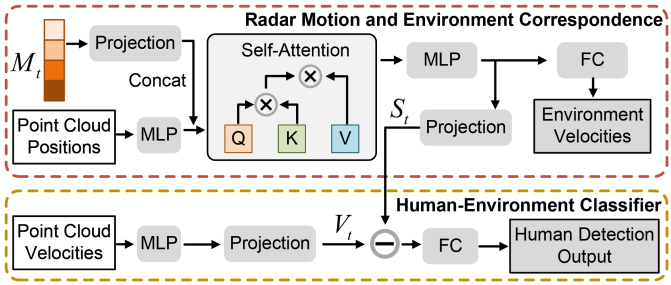


Fig. 10. Model architecture for velocity-aware calibration mechanism.

features. We utilize a fully connected layer without an activation function to perform the projection.

According to (1), the correspondence between the radar's translational velocity and the static environment's radial velocity is dependent on the angle of the environment relative to the radar. However, the environment's velocity and angle refer to the true values rather than the radar's measured values. Due to the characteristics of mmWave radar, the measured angle of a target contains an error relative to the true value, and the error varies with the target's position, where the radar's angular resolution deteriorates as the target's angle increases.

To mitigate the impact of measurement errors on this correspondence, we adopt a non-local operation using self-attention. This allows each point to reference information from other points, thereby avoiding erroneous inferences based on each point's local features. Based on the global features (i.e., F_{global}) of each point including radar motion and point positions, the model uses an MLP to obtain the corresponding static environment's velocity feature at each point position, which is denoted as $F_{static} = MLP(F_{global}; \theta_{static})$, where θ_{static} is the learnable parameters of the MLP. Then the model employs a fully connected layer to obtain the environment velocities.

Loss of Radar Motion and Environment Correspondence: This task is to minimize the error between the predicted radial velocities of static environmental points and corresponding ground truth. And we minimize the MSE loss:

$$L_V = \|\hat{v} - v\|^2, \tag{8}$$

where $\hat{v}$ and v is the predicted radial velocities of static environmental points and corresponding ground truth.

2) Human-Environment Classifier: After the model estimates the radial velocity for a point at a specific location as if it belongs to the static environment, it uses the radar-measured actual velocity of that point to determine whether it belongs to a human or the environment. Specifically, we subtract the measured velocity feature V_t from the model's estimated static environment velocity feature S_t. If the point is a part of the static environment, the two features will be similar, and the difference should be close to zero. Conversely, if the point belongs to a human, the difference will deviate significantly from zero. This ensures that the model distinguishes between human and environmental points based solely on scene-independent velocity features, thereby enhancing its generalization.

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As illustrated in the lower part of Fig. 10, the model first uses an MLP to expand the radar-measured point cloud velocity information. The output is denoted as $F_{vel} = MLP(e^{vel}; \theta_{vel})$, where e^{vel} represents radar points' velocities and θ_{vel} is the learnable parameters of the MLP. Then, the actual measured velocity feature of each point and the estimated static environment velocity feature from the previous component are projected into the same semantic space and subtracted. Based on the result of this subtraction, the model employs a fully connected layer to decide whether the point belongs to a human.

Loss of People Detection: The people detection task is to minimize the error between the predicted class (human or environment) and the ground truth class of every point. And we minimize the Cross Entropy loss:

$$L_C = -\left[y_m \log(p(\hat{y}_m)) + (1 - y_m) \log\left(1 - p(\hat{y}_m)\right)\right], \quad (9)$$

where y_m is the function of 0 and 1, $y_m = 1$ if the class of sample m is human, and vice versa. $p(\hat{y}_m)$ is the prediction probability that sample m belongs to human.

Multi-task Loss: Due to the different values of the losses for each task, there is a risk that tasks with larger loss values dominate the training process. To ensure that all tasks can be sufficiently trained, we automatically adjust the weights of each task's loss based on the training result [40], [45]. The total training loss is:

$$L = \sum_{t \in T} \left[\frac{1}{2 \cdot c_t^2} \cdot L_t + \ln\left(1 + c_t^2\right)\right], \quad (10)$$

where $L_T = \{L_A, L_W, L_V, L_C\}$ is the set of above four task losses. c_t denotes a learnable network parameter, and $\ln(1 + c_t^2)$ represents a regularization term.

V. Implementation

A. Hardware

mmWave Radar: Mobi²Still is implemented with a COTS mmWave radar Texas Instruments (TI) IWR6843ISK [31] for point cloud acquisition. The radar has three transmit antennas and four receive antennas. It generates mmWave chirp signals operating from 60.75 - 64 GHz. The periodicity of frame is 55 ms and there are 288 chirps in each frame. The radar can reach the range resolution of 8.43 cm, and the maximum sensing velocity of 4.62 m/s with a 0.096 m/s resolution.

Femto Bolt Camera: To obtain the ground truth of all tasks, we utilize Femto Bolt Camera [32]. Using its RGB and infrared cameras, the Femto Bolt can capture depth images of the human body, generate skeletal joints, and perform skeleton tracking. We use the identity (ID) and position of each human target from the skeleton tracking, as the ground truth for the people tracking task. For the people detection task, we label mmWave point clouds with skeletal joints. If the distance between a mmWave point and all skeletal joints exceeds 30 cm, the point is labeled as belonging to the static environment; otherwise, it is labeled as a human point. Besides, we rigidly mount the Femto Bolt and the radar together, and utilize the built-in accelerometer and gyroscope to obtain tri-axis acceleration and angular velocity measurements as the ground truth for radar motion estimation task.

B. Software

Data Processing: To obtain precise point clouds, SO-CFAR [46] is employed to effectively suppress noise, and Capon Beamforming [47] is applied to enhance angular resolution. Besides, Mobi²Still employs extended Kalman filter (EKF) [28] for long-term accurate people tracking.

Model configuration and training: The input of the model is a sequence of mmWave point clouds over a period of time. There are 32 frames in one sequence, and 150 points in one frame. We fix the number of radar points per frame primarily to ensure consistent input dimensions for the model, facilitating the batch training and accelerating convergence [48]. The size of the MLP layer after distance encoding is (256,64). In angle encoding, the number of nearest neighbors N is set to 10, and the sizes of the following MLP layers are (16,1) and (256,64). In the scene geometric construction component, the size of MLP is (16,1). The self-attention in environment geometry match has 4 attention heads, with a hidden size of 256. In the velocity-aware calibration mechanism, all the projection operations are implemented by fully connected layers without activation functions. The self-attention has 4 heads and the hidden size of 320. After point cloud velocities, we implement MLP (4,16,64) for people detection. The activation function is ReLU [49].

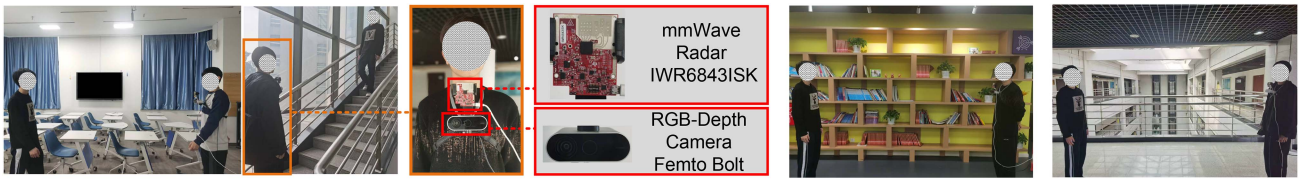
For every activity and subject (details in Section V-C), 80% data is for training and 20% is for testing. The batch size is 8 and the learning rate is 0.0001. We use AdamW [50] as optimizer. Our model is implemented with Python 3.8 and PyTorch 2.4.1, which is trained on a server running Ubuntu 22.04.1 with an NVIDIA RTX 4090 GPU. In terms of inference speed, the average runtime per frame on the NVIDIA Jetson TX2 [51] is 53.4 ms.

C. Data Collection

We recruit 28 volunteers, including 23 males and 5 females, with ages varying from 22 to 29, heights from 157 cm to 188 cm, and weights from 48 kg to 81 kg. All volunteers are fully informed of the purpose of data collection and provided consent for the use of their data in model training and academic publication. We group volunteers into pairs, selecting one as the human equipped with an mmWave radar, and the other as the person performing various actions. In the experiments, the person is instructed to perform four distinct activities that are common in the real world, including *(i)* free walking, *(ii)* free running, *(iii)* bending while walking, and *(iv)* ascending and descending stairs. Note that, these activities are not assigned predetermined routes; instead, the person freely and randomly selects his or her paths. The first three activities are performed on flat surfaces, while the last is conducted on staircases.

To evaluate the system's performance across different scenarios, we collect data in three typical scenarios that include both flat surfaces and staircases. Fig. 11(a) shows a classroom along with an adjacent staircase, and there are many desks and chairs inside the classroom. Fig. 11(b) shows a library with staircases.

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(a) Classroom (b) Library (c) Office building

Fig. 11. Experimental scenarios.

Fig. 11(c) illustrates a corridor within an office building, which has staircases. As shown in Fig. 11(a), we mount the mmWave radar on the upper side of the Femto Bolt and secure both devices to a human using a neck mount and a strap. It is important to note that once the model is trained, camera-based labeling is no longer required. The human equipped with the radar naturally walks on flat surfaces and ascends/descends stairs in the staircase, during which the radar experiences natural rotation and translation. The other person performs the designated activities within an area ranging from 1 to 4 meters away from the human. For each activity in every scenario, each subject is recorded for 5 minutes, during which over 5,400 frames of radar point cloud data are collected.

VI. Evaluation

A. Experimental Setups

Performance metrics: We use the accuracy, precision, recall and F1-score to evaluate the effect of people detection. To evaluate the people tracking task, we utilize the following metrics: *(i)* Multiple Object Tracking Accuracy (MOTA) [52] accounts for detection accuracy and tracking consistency, which is defined as $1 - \frac{\text{FN}+\text{FP}+\text{IDSW}}{\text{GT}}$, where FN represents the number of missed detections; FP denotes the detections of non-existent targets; IDSW is the count of identity switches; GT is the number of true targets. MOTA ranges from negative infinity to 1, with higher values indicating better performance. A negative value indicates extremely poor tracking performance. *(ii)* Multiple Object Tracking Precision (MOTP) [53] measures the spatial matching precision between detection results and true targets, which is defined as $\frac{1}{N}\sum_{i=1}^{N} d(\text{det}_i, \text{gt}_i)$, where $d(\text{det}_i, \text{gt}_i)$ represents the Euclidean distance between the i-th correctly matched detection and its corresponding ground truth; N is the number of correctly matched pairs.

Comparison baselines: (i) RF-Search [20] is a novel approach to detect and track static people from environmental objects using a purely translating mmWave radar. By computing the Pearson correlation coefficients of the motion relative to the radar of all tracked objects, RF-Search identifies the object with the smallest coefficient as the human. In our application scenario, people are moving, and we track all objects based on radar point clouds and use the tracked trajectories as the motion, and name it RF-Search*. *(ii)* RANSAC-LSQ [18] is a classical method widely adopted in the automotive industry for separating moving objects from static environments using mmWave radar, where the radar undergoes translation with minimal rotation.

TABLE I
Overall Performance of Mobi^{2}Still

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MOTA	MOTP (cm)
RF-Search*	66.64	64.48	61.02	60.93	-0.0889	38.68
RANSAC-LSQ	61.49	58.09	57.14	57.11	-1.5011	38.60
RaTrack	91.75	90.07	91.12	90.56	0.8712	30.83
Mobi^{2}Still	**98.36**	**98.09**	**98.10**	**98.10**	**0.9832**	**28.34**

By modeling the motion of the environment and the radar, RANSAC-LSQ detects moving humans from the environment. To ensure a fair comparison with the tracking performance of Mobi^{2}Still, the EKF is employed as the tracking method. *(iii)* RaTrack [19] leverages an end-to-end, multi-task learning model to detect moving targets and associate them with previous detections for continuous tracking.

B. Overall Performance

Mobi^{2}Still achieves high performance for people detection and tracking with human-equipped mmWave radars: As shown in Table I, for the overall datasets, Mobi^{2}Still achieves the performance with an accuracy of 98.36%, precision of 98.09%, recall of 98.10%, F1-score of 98.10%, MOTA of 0.9832, and MOTP of 28.34 cm, outperforming the baselines by at least 6.61%, 8.02%, 6.98%, 7.54%, 11.20% and 2.49 cm, respectively. From these results, we can observe that: *(i)* Mobi^{2}Still achieves the highest performance, demonstrating its effectiveness, which estimates radar motion by matching recurring environment and subsequently enables accurate human detection and tracking based on the inferred motion; *(ii)* The baselines perform poorly because, when the radar is equipped on a human, the frequent rotations result in drastic and unpredictable changes in object positions. This makes it difficult to obtain consistent motion of each object over time, which is essential for these methods to function properly. Mobi^{2}Still overcomes this problem with the help of matching hard-to-find recurring environment by geometric structures. As the geometric structure describes the relative spatial relationships of objects, it remains unaffected by radar rotations.

Fig. 12 demonstrates examples of people detection and tracking. We depict mmWave points using different colors, where blue indicates environmental points and non-blue points represent human-related points. Human targets are tracked based on the human points. Points belonging to different human targets are represented using distinct colors, while noise points that are not

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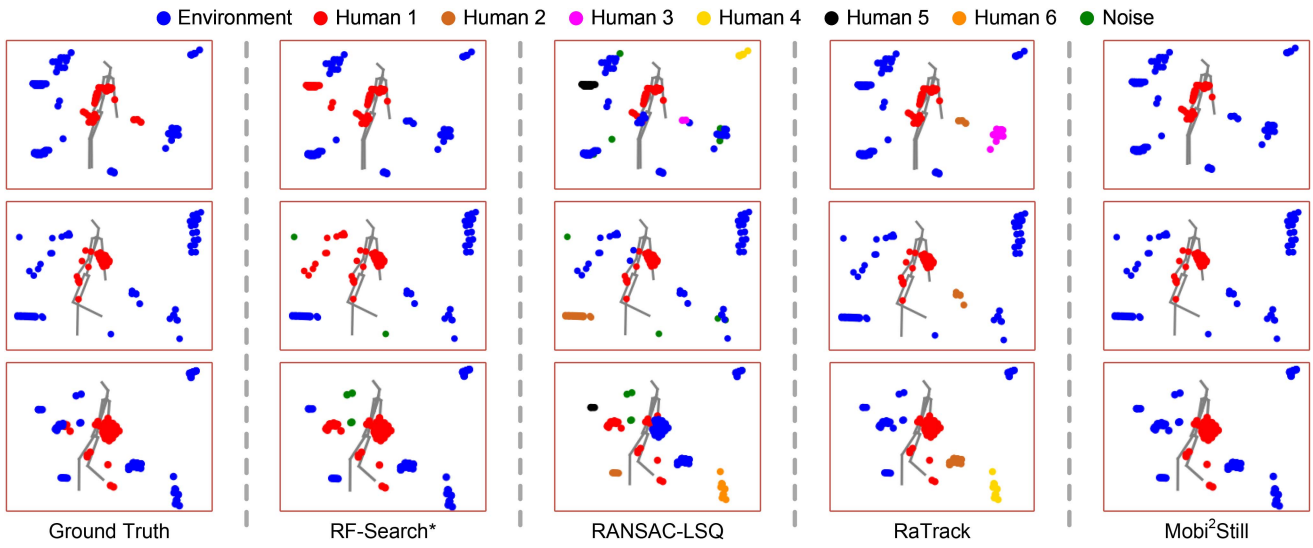


Fig. 12. Qualitative results.

considered as a human are shown in green. The gray skeleton in each figure is generated by the camera and only for visualization. The three rows correspond to a continuous action of ascending and descending stairs. The first column shows the corresponding ground truth.

We can find that: *(i)* The last column shows the results of Mobi²Still. Despite the continuous rotation and translation of human-equipped radar and the hard-to-find recurring environments, our system is still able to detect and track the person accurately; *(ii)* Compared with Mobi²Still, the baseline methods mistakenly identify parts of the environmental points as the human, and parts of the human points as the environment, leading to inaccurate and erroneous tracking. The improvements are achieved by the design of extracting and matching geometric structures of objects. The geometric structures are stable when radar undergoes translation and frequent rotation and help the matching of recurring environment based on the geometry similarity, ensuring effective people detection and tracking.

Mobi²Still achieves generalization in unknown scenes: We implement the scene-independent experiments, including a people-independent experiment, where the test participants (6 subjects) have not been included in the model training datasets (the other 22 subjects), and an environment-independent experiment where we sequentially select two environments from the three as the training set and use the remaining unseen environment as the test set. We use macro-averaging to calculate the arithmetic mean of the metrics for each unseen environment.

Table II shows the accuracy, F1-score, and MOTA for unseen people and environments, respectively. We can observe that, *(i)* for unseen people, Mobi²Still achieves 96.21% accuracy, 95.70% F1-score, and 94.37% MOTA, reflecting reductions of 2.15%, 2.40% and 3.95%, respectively, compared with the results on seen individuals. In contrast, baselines perform poorly, with accuracy below 0.85, F1-score below 0.81, and MOTA below 0.71. For unseen environments, Mobi²Still attains 97.08% accuracy, 96.68% F1-score, and 97.63% MOTA, corresponding

TABLE II
PERFORMANCE OF UNSEEN PEOPLE AND ENVIRONMENTS

Method	Unseen People			Unseen Environments		
	Accuracy	F1-score	MOTA	Accuracy	F1-score	MOTA
RF-Search*	0.6225	0.5555	-0.2300	0.6605	0.5967	-0.1306
RANSAC-LSQ	0.6252	0.5867	-1.5137	0.6285	0.5865	-1.4968
RaTrack	0.8417	0.8087	0.7045	0.8132	0.7761	0.5156
Mobi²Still	**0.9621**	**0.9570**	**0.9437**	**0.9708**	**0.9668**	**0.9763**

to decreases of 1.28%, 1.42%, and 0.69%, respectively, relative to seen environments. Baseline methods exhibit inferior results, with accuracy below 0.82, F1-score below 0.78, and MOTA below 0.52. All these results show the strong generalization of Mobi²Still in unknown scenes. The advantage comes from our design of a velocity-aware calibration mechanism in people detection, which establishes the detection rule using scene-independent velocity information. Although the network parameters are not fine-tuned by the test data, the mechanism can establish the radar motion and static environment correspondence on velocity and use it to guide the people detection process. *(ii)* For RF-Search* and RANSAC-LSQ, there exists the phenomenon of the scene-independent performance being higher than the overall performance, which is due to the fact that both methods are rule-based, non-neural methods. Since all the collected data is unseen to them, their performance can fluctuate to some extent across different datasets.

To intuitively explain the generalization of Mobi²Still, we visualize the feature representations of each mmWave point before the fully connected layer of the human-environment classifier in Fig. 13. These points belong to different humans and environments across different scenes, including S1 (classroom), S2 (library), and S3 (office building). We apply t-SNE [36] to project the high-dimensional features into a two-dimensional space. Each point in the figure corresponds to a feature of a mmWave point labeled as either human or environment. Unlike Fig. 5, where scene-related features lead to clustering by

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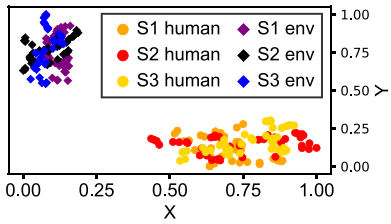


Fig. 13. The visualization for the environmental and human features extracted by Mobi^{2}Still from different scenes.

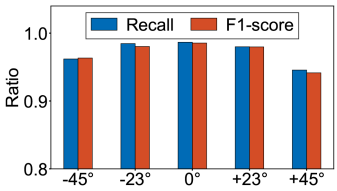


Fig. 14. Evaluation at different view angles. - means the left side; + means the right side.

scene, our model not only clearly separates human and environmental features, but also makes the features from different scenes uniformly distributed within their respective classes, indicating that the extracted representations are invariant to the scene. This demonstrates that our model maintains strong performance to unseen individuals and environments, due to the design of the velocity-aware calibration mechanism to extract scene-independent velocity.

C. Performance Under Different Conditions

Impact of different view angles: We evaluate the performance against various view angles between the radar-equipped human and the target subject. To simulate different view angles, we instruct the human to walk along a straight path, while subjects are asked to perform activities at varying offsets from this path. This setup results in relative view angles between the subject and the radar-equipped human of 0°, ± 23°, and ± 45°. As shown in Fig. 14, we can observe that, *(i)* Mobi^{2}Still achieves stable performance with 94.56% recall and 94.15% F1-score at a very large angle +45°, showing that our system has good robustness when the angle changes. *(ii)* The performance of our model decreases as the angle increases because the angular resolution degrades with increasing angle, leading to a certain degree of disruption in the correspondence between radar motion and the environment.

Impact of different distances: In realistic scenarios, the distance between the radar-equipped human and the target subject may vary. To assess the impact of distance on system performance, we conduct experiments at distances ranging from 1 m to 5 m. As shown in Fig. 15, *(i)* performance remains stable between 1 m and 4 m, where the system achieves 95.72% precision and 96.07% F1-score at 4 m. *(ii)* However, at 5 m, we observe a degradation (91.81% precision and 92.34% F1-score), primarily attributed to the limited radar power. This limitation can be mitigated by employing a higher-power radar.

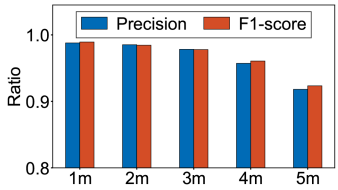


Fig. 15. Evaluation at different distances between target subject and radar-equipped human.

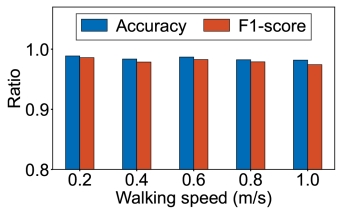


Fig. 16. Evaluation of different walking speeds of a radar-equipped human.

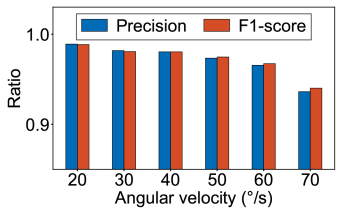


Fig. 17. Performance of different angular velocities.

Impact of different walking speeds of the radar-equipped human: In real-world scenarios, humans can walk at different speeds. To assess the impact of different walking speeds of the radar-equipped human on system performance, we conduct experiments with the human moving at speeds ranging from 0.1 m/s to 1.0 m/s. As illustrated in Fig. 16, the system maintains strong performance across varying walking speeds, achieving an accuracy of 98.18% and an F1-score of 97.42% at the speed of 1.0 m/s, demonstrating its adaptability to real-world scenarios.

Impact of different angular velocities: In this subsection, we investigate whether the angular velocities of the radar influence the performance of our system. Specifically, we conduct experiments with the radar rotating at average angular velocities ranging from 20 °/s to 70 °/s. As shown in Fig. 17, *(i)* the system maintains strong performance between 20 °/s and 60 °/s (96.55% precision and 96.73% F1-score at 60 °/s). *(ii)* However, at 70 °/s, we observe a noticeable performance drop (93.62% precision and 94.01% F1-score). This degradation is attributed to the reduced recurring environment at higher angular velocities, which impairs the accuracy of radar motion estimation and affects the performance of Mobi^{2}Still.

Impact of different radar types: We next investigate whether the radar types influence the system performance. Specifically, we conduct experiments using three different radar hardware platforms, including TI IWR6843ISK, IWR6843ISK-ODS [54], and IWR1443BOOST [55]. These radar devices differ in their operating frequencies or antenna array layouts. In particular,

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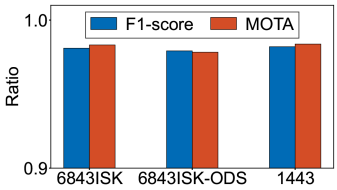


Fig. 18. Evaluation of different radar types.

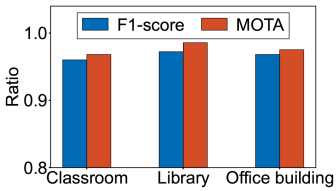


Fig. 19. Performance of different environments.



(a) Straight paths (b) "Z" paths

Fig. 20. Evaluation of different paths of a radar-equipped human.

TABLE III
PERFORMANCE OF DIFFERENT ACTIVITIES

Activity	Accuracy	MOTA
Walking	0.9896	0.9890
Running	0.9844	0.9795
Bending over	0.9882	0.9839
Up/downstairs	0.9622	0.9722

the operating frequency of the IWR6843ISK and IWR6843ISK-ODS is 60 GHz, while that of the IWR1443BOOST is 77 GHz. The number of horizontal virtual antenna channels is 8 for both the IWR6843ISK and IWR1443BOOST, and 4 for the IWR6843ISK-ODS. As shown in Fig. 18, *(i)* the system maintains strong performance across all three radars (97.92% F1-score and 97.83% MOTA with IWR6843ISK-ODS), demonstrating the generalization of the proposed geometric structure based recurring environment matching and velocity-aware calibration mechanism of Mobi²Still across different radar devices. *(ii)* The system exhibits relatively lower performance on the IWR6843ISK-ODS, as its antenna layout leads to lower azimuth angular resolution compared to the other radar hardware, resulting in poorer point cloud quality.

Impact of different environments: We evaluate the performance of the system in three distinct unseen environments, including classroom, library, and office building. As shown in Fig. 19, *(i)* the system achieves high performance in people detection and tracking across all environments (96.01% F1-score and 96.81% MOTA in classroom), demonstrating its adaptability under varying environmental conditions, which can be attributed to the proposed velocity-aware calibration mechanism. *(ii)* The relatively lower performance in classroom may be due to the close distance between target subjects and environmental objects such as desks and chairs, which leads to some environmental points being mistakenly labeled as human, thereby affecting the model's performance.

Impact of different paths of a radar-equipped human: In real-world scenarios, the radar-equipped human may follow various paths depending on the specific situation. As illustrated in Fig. 20(a) and (b), the human follows three straight paths and two Z shape paths, and the target subject is instructed to walk in a straight line in the direction he or she is facing. *(i)* For the five paths, the F1-score are 98.54%, 98.12%, 95.70%, 97.57% and 97.09%, respectively, demonstrating a high performance under different moving paths. *(ii)* The relatively lower performance in path 3 may be due to the radar and target movement paths being perpendicular, causing the radial velocity of some target points to be zero at certain moments, leading them to be mistakenly classified as the environment.

Impact of different activities: As illustrated in Table III, Mobi²Still achieves high performance across different activities, with even the lowest-performing up/downstairs, reaching 96.22% accuracy and 97.22% MOTA, demonstrating its robustness under different activities. A slight drop is observed in the up/downstairs activity, which is because the subject is very close to the environment such as railings and walls in the stair, leading to some environmental points being labeled as human, which negatively affects model performance.

D. Micro-Benchmark Evaluation

Effectiveness of geometric structure: To match recurring environment, we propose to extract the geometric structures of scene objects and match based on geometry similarity. To validate the effectiveness of geometric structure, which is extracted through distance encoding and angle encoding. We conduct experiments with the following 4 different settings: *(i)* Full version (Full), all the design components are used; *(ii)* W/O encoding, the branches of distance encoding and angle encoding are disabled; *(iii)* Only distance encoding (disEncoding), the branch of angle encoding is disabled, where the model extracts geometric structure only based on distance information; *(iv)* Only angle encoding (angleEncoding), the branch of distance encoding is disabled, and geometric structure is extracted only by angle relationship.

As shown in Fig. 21, we can observe that, *(i)* the full version delivers the highest performance, demonstrating that both angle encoding and distance encoding are useful for the overall people detection and tracking task. *(ii)* When using only angle encoding, the F1-score only drops by 4.19% and MOTA by 5.47% compared to the full version, showing better performance than using only distance encoding. Since distance describes the proximity between objects while angle reflects their relative direction, the result suggests that relative direction plays a more significant role in the geometric structure. *(iii)* Compared to the w/o encoding setting, using only distance encoding leads to

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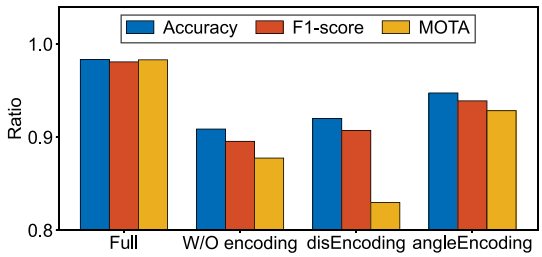
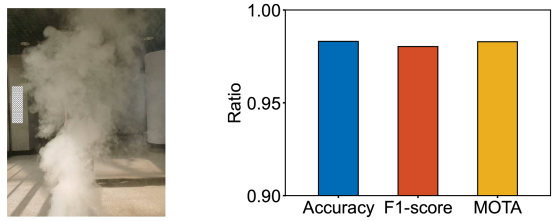


Fig. 21. The impact of geometric structure on people detection and tracking.



(a) Smoke scenario (b) Performance of Mobi^{2}Still

Fig. 22. Evaluation of Mobi^{2}Still under smoke scenario.

TABLE IV
THE IMPACT OF VELOCITY-AWARE CALIBRATION

Models	Seen environments (%)			Unseen environments (%)		
	Accuracy	F1-score	MOTA	Accuracy	F1-score	MOTA
CNN	97.25	96.96	95.42	87.66	84.54	78.28
ResNet	97.32	97.03	96.81	87.51	83.95	81.81
PointNet	98.04	97.83	97.32	85.38	83.58	72.87
PointNet+LSTM	98.35	98.17	96.95	92.97	91.47	86.73
Mobi^{2}Still *	98.21	98.00	97.04	93.12	92.07	88.74
Mobi^{2}Still	**98.14**	**97.83**	**98.30**	**97.08**	**96.68**	**97.63**

improved accuracy and F1-score, but results in a drop in MOTA. The reason is that distance captures the proximity between objects, offering a simple spatial relationship that enhances people detection performance. However, this may cause the model to overemphasize proximity as the primary cue for identifying human points, leading it to mistakenly classify clusters of closely located environmental points as human, thus affecting tracking performance.

Impact of velocity-aware calibration: As mentioned in Section IV-B, to enable Mobi^{2}Still to achieve good generalization in unseen scenes, we utilize a velocity-aware calibration mechanism to force the model to focus on scene-independent velocity as the basis for detecting people from the environment. To evaluate the impact of the velocity-aware calibration mechanism, we implement 5 different models without the calibration process. Specifically, *(i)* we replace the calibration mechanism with classic neural network Convolutional Neural Network (CNN) [56], Residual Network (ResNet) [39], PointNet [57] and PointNet with LSTM [58], respectively. *(ii)* We discard the environment velocities estimation and the feature subtraction in the calibration mechanism to disable its ability related to scene-independent velocity, which is named Mobi^{2}Still *. In this ablation study, we sequentially select two environments from the three for training, and use the remaining unseen environment for test.

As shown in Table IV, we observe that, *(i)* Mobi^{2}Still achieves 97.08% accuracy, 96.68% F1-score, and 97.63% MOTA in unseen environments, outperforming the baselines by at least 3.96%, 4.61%, and 8.89%, respectively, showing the effectiveness of velocity-aware calibration mechanism for model's generalization capability. *(ii)* The reason that PointNet with LSTM and Mobi^{2}Still * slightly outperforms Mobi^{2}Still in seen environments is that they falsely rely on scene-specific information for people detection. Although this leads to better performance in seen scenes, it negatively affects their generalization in unseen scenes.

Performance of auxiliary multi-tasks: To evaluate the performance of auxiliary multi-tasks, we adopt mean squared error (MSE), which is calculated by the mean of the squared errors between the predicted values and the actual values. *(i)* The MSE of radar acceleration and angular velocity estimation tasks are 0.37 and 0.03, respectively, indicating Mobi^{2}Still can obtain accurate radar motion based on the design of matching recurring environments using geometric structures. *(ii)* The MSE of environmental velocities estimation for radar motion and environment correspondence task is 0.18, showing that Mobi^{2}Still successfully establishes the correspondence in velocity dimension, thus improving the generalization.

E. Case Study in the Extreme Smoke Scenario

As illustrated in Fig. 22(a), we use a smoke generator to produce a dense smoke environment to test Mobi^{2}Still. Due to occlusion caused by the smoke, the visual camera fails to construct ground truth for people detection and tracking. Therefore, we use an AirTag [59] as a substitute. Specifically, the AirTag is attached to the target's chest, and the iPhone is mounted together with the mmWave radar to obtain the target's position relative to the radar, which is utilized to assist manual ground truth annotation. In addition, we note that this dense smoke setup complies with our local fire safety regulations and other experimental settings remain identical to those used in the smoke-free condition.

As shown in Fig. 22(b), Mobi^{2}Still achieves 98.31% accuracy, 98.03% F1-score, and 98.29% MOTA, indicating that its performance in smoke-filled environments is comparable to that in smoke-free scenarios. This demonstrates that Mobi^{2}Still can operate reliably under dense smoke, highlighting its strong potential for real-world applications.

VII. RELATED WORK

Wireless sensing leverages various RF signals including Wi-Fi [60], [61], LTE [62], [63], LoRa [64], [65], RFID [66], [67], UWB [68], [69], and mmWave [1], [16], [70], [71] to understand the surrounding world. Among these signals, mmWave stands out for its fine-grained sensing capability, offering excellent velocity, range, and angular resolution enabled by its high frequency, large frequency bandwidth, and multi-antenna array. Therefore, mmWave is adopted in various sensing applications,

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including fall detection [70], [72], [73], gesture recognition [70], [74], [75], vital sign monitoring [70], [76], [77], [78], emotion and mental stress inference [71], [79], [80], and people detection and tracking [1], [15], [16]. In this work, we adopt mmWave radar as the sensing device due to its ability to provide high-resolution velocity, range and angle information, which facilitates the separation of human reflections from environmental objects and achieves high precision of people tracking.

People detection and tracking using mmWave: Recent years have witnessed significant advancements in mmWave-based people detection and tracking [1], [13], [14], [15], [16]. For instance, Waffle [16] accurately detects and tracks people in the bathroom, and proposes a customized Constant False Alarm Rate detector to eliminate the interference from running water. Chen et al. [15] estimate the locations of static environmental objects from received mmWave signals, to eliminate the negative multi-path interference. PmTrack [1] develops a personalized mmWave-based people tracking system utilizing IMUs as identity indicators. However, most mmWave-based people detection and tracking systems assume that the mmWave radar keeps stationary during the sensing procedure. In this paper, we aim to perform people detection and tracking when the radar is in frequent motion, enhancing its applicability in dynamic scenarios.

Wireless sensing with moving platforms: Recent studies have increasingly focused on wireless sensing with various moving platforms, including vehicles [17], [18], [19], drones [20], wheeled robots [21], [22] and human hands [23], [24], [25]. In vehicle platforms, MSense [17] equips the sensing device in the driver cabin and achieves eye blinking detection, by using adjacent body regions of the target body part as reference points to suppress both device and irrelevant body motion. RANSAC-LSQ [18] detects moving people from the environment by modeling the motion of the environment and sensing device. RaTrack [19] detects and tracks moving targets through multi-task learning that integrates motion segmentation and affinity matrix computation. Drone-based approach, RF-Search [20], compares motion similarity across all objects, including environment and humans, and identifies the one with the lowest similarity as the human. In the wheeled robot platforms, RobotSen [21] leverages mobile LoRa receivers to extend the sensing range. Mobi²Sense [22] utilizes static objects in the environment as references to cancel out radar motion from human signals. In human hand platforms, Liu et al. [23] propose a dynamic Fresnel zone model for Wi-Fi sensing, and enable gesture recognition. M²-Fi [24] enables respiration monitoring with handheld Wi-Fi receivers, and employs Variational Mode Decomposition to eliminate the interference caused by hand movement. MmECare [25] introduces two corner reflectors near the human body as references to monitor breathing and heartbeat. Although these methods achieve good performance with their moving platforms, where sensing devices primarily translate with minimal rotation, none of them have addressed wireless sensing with human-equipped devices, where the device simultaneously translates and rotates. Mobi²Still enhances the sensing capabilities when the device is mounted on a human, by proposing a geometric structure-based recurring environment match method to ignore the effects of frequent device rotation and translation.

Radar point cloud alignment, which aims to match corresponding points captured at different times, has been widely applied in radar-based simultaneous localization and mapping (SLAM). Recent studies have achieved significant progress on vehicles [81], [82], [83], wheeled robots [42], [84], and drone platforms [41]. In vehicle platforms, Doppler Correspondence [81] directly aligns point clouds based on each point's range and doppler velocity, eliminating the need for iterative optimization. Generalized-ICP [82] models locally planar surface structure from two different point clouds. Herraez et al. [83] utilize the doppler velocity information to improve the point-to-point iterative closest point (ICP) algorithm. In wheeled robot platforms, 4DRadarSLAM [42] takes into account the spatial probability distribution of each point in Generalized-ICP. Xu et al. [84] model and incorporate the uncertainty of radar points to enhance performance. In drone platforms, Doer et al. [41] propose an Extended Kalman Filter based method and reduce the computational burden. Despite their effectiveness on corresponding moving platforms, most of which primarily translate with minimal rotational motion, existing approaches are not applicable to human-equipped radars, where translation and rapid rotation occur simultaneously. Mobi²Still utilizes the geometric structures of objects, which capture invariant spatial relationships including relative distances, directions, and angles, to suppress the influence of frequent rotation and translation, thereby enabling effective people detection and tracking.

VIII. Discussion

We next discuss the future direction of Mobi²Still.

Extending the detection range: Although Mobi²Still significantly outperforms existing methods, its performance still degrades when the distance between the target person and the radar-equipped human exceeds 4 meters, as evaluated in Section VI-C. This limitation arises from the constrained transmission power of the radar. In future work, we will upgrade to higher-power radar hardware to extend the effective detection range, further enhancing the performance of Mobi²Still.

Validation in a more diverse population: This paper primarily involves participants recruited from a university, with a limited age range focused on young and early-middle-aged healthy adults. In future research, we plan to expand the participant pool to include a wider age spectrum, such as children and elderly individuals, as well as those with health conditions, to enhance Mobi²Still's robustness.

IX. Conclusion

To fill the gap of mobile wireless sensing in human-worn scenarios, we propose Mobi²Still, a novel human-equipped mmWave sensing system for people detection and tracking. To eliminate the influence of radar translation and frequent rotation, we design a geometric structure based recurring environment matching method. To enhance Mobi²Still's generalization, we then introduce a velocity-aware calibration mechanism that

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focuses on scene-independent features. Extensive experiments demonstrate that Mobi^{2}Still achieves 98.10% F1-score and 98.32% MOTA, outperforming the state-of-the-art methods by at least 7.54% F1-score and 11.20% MOTA.

REFERENCES

[1] H. Liu, X. Liu, X. Xie, X. Tong, and K. Li, "PMTrack: Enabling personalized mmWave-based human tracking," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 7, no. 4, pp. 1–30, 2024.

[2] B. Amirgaliyev, M. Mussabek, T. Rakhimzhanova, and A. Zhumadillayeva, "A review of machine learning and deep learning methods for person detection, tracking and identification, and face recognition with applications," *Sensors*, vol. 25, no. 5, 2025, Art. no. 1410.

[3] A. Pallaprolu, P. Peng, S. Sandhu, W. Hurst, and Y. Mostofi, "Crowd analytics with a single mmWave radar," in *Proc. 30th Annu. Int. Conf. Mobile Comput. Netw.*, 2024, pp. 830–844.

[4] Z.-Q. Cheng, Q. Dai, H. Li, J. Song, X. Wu, and A. G. Hauptmann, "Rethinking spatial invariance of convolutional networks for object counting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 19638–19648.

[5] S. Yue, H. He, P. Cao, K. Zha, M. Koizumi, and D. Katabi, "CornerRadar: RF-based indoor localization around corners," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 6, no. 1, pp. 1–24, 2022.

[6] M. Pervaiz, A. Jalal, and K. Kim, "Hybrid algorithm for multi people counting and tracking for smart surveillance," in *Proc. Int. Bhurban Conf. Appl. Sci. Technol.*, 2021, pp. 530–535.

[7] K. Lee, D. Sato, S. Asakawa, H. Kacorri, and C. Asakawa, "Pedestrian detection with wearable cameras for the blind: A two-way perspective," in *Proc. CHI Conf. Hum. Factors Comput. Syst.*, 2020, pp. 1–12.

[8] O. H. Jafari, D. Mitzel, and B. Leibe, "Real-time RGB-D based people detection and tracking for mobile robots and head-worn cameras," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2014, pp. 5636–5643.

[9] J. Liu, Y. Liu, G. Zhang, P. Zhu, and Y. Q. Chen, "Detecting and tracking people in real time with RGB-D camera," *Pattern Recognit. Lett.*, vol. 53, pp. 16–23, 2015.

[10] T. Taipalus and J. Ahtiainen, "Human detection and tracking with knee-high mobile 2D LiDAR," in *Proc. IEEE Int. Conf. Robot. Biomimetics*, 2011, pp. 1672–1677.

[11] J. Gómez, O. Aycard, and J. Baber, "Efficient detection and tracking of human using 3D LiDAR sensor," *Sensors*, vol. 23, no. 10, 2023, Art. no. 4720.

[12] C. Álvarez-Aparicio, A. M. Guerrero-Higueras, F. J. Rodrıguez-Lera, J. Ginés Clavero, F. Martıń Rico, and V. Matellán, "People detection and tracking using LiDAR sensors," *Robotics*, vol. 8, no. 3, 2019, Art. no. 75.

[13] J. Pegoraro, F. Meneghello, and M. Rossi, "Multiperson continuous tracking and identification from mm-Wave micro-Doppler signatures," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 2994–3009, Apr. 2021.

[14] C. Wu, F. Zhang, B. Wang, and K. R. Liu, "mmTrack: Passive multi-person localization using commodity millimeter wave radio," in *Proc. IEEE Conf. Comput. Commun.*, 2020, pp. 2400–2409.

[15] W. Chen et al., "Environment-aware multi-person tracking in indoor environments with mmWave radars," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 7, no. 3, pp. 1–29, 2023.

[16] X. Zhang, D. Zhang, Y. Xie, D. Wu, Y. Li, and D. Zhang, "Waffle: A waterproof mmWave-based human sensing system inside bathrooms with running water," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 7, no. 4, pp. 1–29, 2024.

[17] Z. Chang, F. Zhang, J. Xiong, W. Chen, and D. Zhang, "MSense: Boosting wireless sensing capability under motion interference," in *Proc. 30th Annu. Int. Conf. Mobile Comput. Netw.*, 2024, pp. 108–123.

[18] D. Kellner, M. Barjenbruch, J. Klappstein, J. Dickmann, and K. Dietmayer, "Instantaneous ego-motion estimation using Doppler radar," in *Proc. 16th Int. IEEE Conf. Intell. Transp. Syst.*, 2013, pp. 869–874.

[19] Z. Pan, F. Ding, H. Zhong, and C. X. Lu, "RaTrack: Moving object detection and tracking with 4D radar point cloud," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2024, pp. 4480–4487.

[20] B.-B. Zhang, D. Zhang, R. Song, B. Wang, Y. Hu, and Y. Chen, "RF-search: Searching unconscious victim in smoke scenes with RF-enabled drone," in *Proc. 29th Annu. Int. Conf. Mobile Comput. Netw.*, 2023, pp. 1–15.

[21] B. Xie, D. Ganesan, and J. Xiong, "Embracing LoRA sensing with device mobility," in *Proc. 20th ACM Conf. Embedded Netw. Sensor Syst.*, 2022, pp. 349–361.

[22] F. Zhang, J. Xiong, Z. Chang, J. Ma, and D. Zhang, "Mobi2Sense: Empowering wireless sensing with mobility," in *Proc. 28th Annu. Int. Conf. Mobile Comput. Netw.*, 2022, pp. 268–281.

[23] J. Liu et al., "Towards a dynamic fresnel zone model to WiFi-based human activity recognition," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 7, no. 2, pp. 1–24, 2023.

[24] J. Hu et al., "M 2-Fi: Multi-person respiration monitoring via handheld WiFi devices," in *Proc. IEEE Conf. Comput. Commun.*, 2024, pp. 1221–1230.

[25] Z. Chang et al., "MmECare: Enabling fine-grained vital sign monitoring for emergency care with handheld mmWave radars," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 8, no. 4, pp. 1–24, 2024.

[26] S. M. Bruijn, O. G. Meijer, J. H. Van Dieen, I. Kingma, and C. J. Lamoth, "Coordination of leg swing, thorax rotations, and pelvis rotations during gait: The organisation of total body angular momentum," *Gait Posture*, vol. 27, no. 3, pp. 455–462, 2008.

[27] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Eng.*, vol. 82, pp. 35–45, 1960.

[28] M. I. Ribeiro, "Kalman and extended Kalman filters: Concept, derivation and properties," *Inst. Syst. Robot.*, vol. 43, no. 46, pp. 3736–3741, 2004.

[29] N. J. Gordon, D. J. Salmond, and A. F. Smith, "Novel approach to nonlinear/non-Gaussian Bayesian state estimation," *IEE Proc. F (Radar Signal Process.)*, vol. 140, no. 2, pp. 107–113, 1993.

[30] P. Beckmann and A. Spizzichino, *The Scattering of Electromagnetic Waves From Rough Surfaces*. Norwood, MA, USA: Artech House, 1987.

[31] Texas instruments iwr6843isk, 2025. Accessed: Apr. 24, 2025. [Online]. Available: https://www.ti.com/tool/IWR6843ISK

[32] Orbbec femto bolt, 2025, Accessed: Apr. 24, 2025. [Online]. Available: https://www.orbbec.com/products/tof-camera/femto-bolt/

[33] E. T. Whittaker, "A Treatise on the Anal. Dynamics of Particles and Rigid Bodies.," Cambridge, U.K.: Cambridge Univ. Press, 1964.

[34] N. El-Sheimy and A. Youssef, "Inertial sensors technologies for navigation applications: State of the art and future trends," *Satell. Navigation*, vol. 1, no. 1, 2020, Art. no. 2.

[35] H. Abdi and L. J. Williams, "Principal component analysis," *Wiley Interdiscipl. Rev.: Comput. Statist.*, vol. 2, no. 4, pp. 433–459, 2010.

[36] B. Melit Devassy, S. George, and P. Nussbaum, "Unsupervised clustering of hyperspectral paper data using T-SNE," *J. Imag.*, vol. 6, no. 5, 2020, Art. no. 29.

[37] F. Rosenblatt, "The perceptron: A probabilistic model for information storage and organization in the brain," *Psychol. Rev.*, vol. 65, no. 6, 1958, Art. no. 386.

[38] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.

[39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.

[40] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7482–7491.

[41] C. Doer and G. F. Trommer, "An EKF based approach to radar inertial odometry," in *Proc. IEEE Int. Conf. Multisensor Fusion Integr. Intell. Syst.*, 2020, pp. 152–159.

[42] J. Zhang et al., "4DRadarSLAM: A 4D imaging radar slam system for large-scale environments based on pose graph optimization," in *Proc. 2023 IEEE Int. Conf. Robot. Automat.*, 2023, pp. 8333–8340.

[43] Q. Huang, Y. Liang, Z. Qiao, S. Shen, and H. Yin, "Less is more: Physical-enhanced radar-inertial odometry," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2024, pp. 15966–15972.

[44] D. Kellner, M. Barjenbruch, J. Klappstein, J. Dickmann, and K. Dietmayer, "Instantaneous ego-motion estimation using multiple Doppler radars," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2014, pp. 1592–1597.

[45] L. Liebel and M. Körner, "Auxiliary tasks in multi-task learning," 2018, *arXiv:1805.06334*.

[46] P. P. Gandhi and S. A. Kassam, "Analysis of CFAR processors in non-homogeneous background," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 24, no. 4, pp. 427–445, Jul. 1988.

[47] J. Capon, "High-resolution frequency-wavenumber spectrum analysis," *Proc. IEEE*, vol. 57, no. 8, pp. 1408–1418, Aug. 1969.

[48] "Batch training," 2025. Accessed: Nov. 28, 2025. [Online]. Available: https://docs.nvidia.com/deeplearning/tensorrt/latest/performance/best-practices.html#batching

[49] Y. Li and Y. Yuan, "Convergence analysis of two-layer neural networks with ReLU activation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 597–607.

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[50] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.
[51] Nvidia jetson tx2 module, 2025. Accessed: Nov. 5, 2025. [Online]. Available: https://developer.nvidia.com/embedded/jetson-tx2
[52] K. Bernardin, A. Elbs, and R. Stiefelhagen, "Multiple object tracking performance metrics and evaluation in a smart room environment," in *Proc. 6th IEEE Int. Workshop Vis. Surveill. Conjunction ECCV*, 2006, pp. 1–8.
[53] K. Bernardin and R. Stiefelhagen, "Evaluating multiple object tracking performance: The clear MOT metrics," *EURASIP J. Image Video Process.*, vol. 2008, pp. 1–10, 2008.
[54] "Texas instruments iwr6843isk-ods," 2025. Accessed: Nov. 4, 2025. [Online]. Available: https://www.ti.com/tool/IWR6843ISK-ODS
[55] "Texas instruments iwr1443boost," 2025. Accessed: Nov. 4, 2025. [Online]. Available: https://www.ti.com/tool/IWR1443BOOST
[56] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
[57] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 652–660.
[58] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
[59] Airtag, 2025. Accessed: Jul. 31, 2025. [Online]. Available: https://www.apple.com/airtag/
[60] W. Li et al., "WiFi-CSI difference paradigm: Achieving efficient doppler speed estimation for passive tracking," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 8, no. 2, pp. 1–29, 2024.
[61] Y. Li et al., "DiverSense: Maximizing Wi-Fi sensing range leveraging signal diversity," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 6, no. 2, pp. 1–28, 2022.
[62] Y. Feng, Y. Xie, D. Ganesan, and J. Xiong, "LTE-based pervasive sensing across indoor and outdoor," in *Proc. 19th ACM Conf. Embedded Netw. Sensor Syst.*, 2021, pp. 138–151.
[63] Y. Feng, Y. Xie, D. Ganesan, and J. Xiong, "LTE-based low-cost and low-power soil moisture sensing," in *Proc. 20th ACM Conf. Embedded Netw. Sensor Syst.*, 2022, pp. 421–434.
[64] B. Xie, M. Cui, D. Ganesan, and J. Xiong, "Wall matters: Rethinking the effect of wall for wireless sensing," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 7, no. 4, pp. 1–22, 2024.
[65] B. Xie, M. Cui, D. Ganesan, X. Chen, and J. Xiong, "Boosting the long range sensing potential of LoRA," in *Proc. 21st Annu. Int. Conf. Mobile Syst. Appl. Serv.*, 2023, pp. 177–190.
[66] Z. Sun, T. Ni, Y. Chen, D. Duan, K. Liu, and W. Xu, "RF-EGG: An RF solution for fine-grained multi-target and multi-task EGG incubation sensing," in *Proc. 30th Annu. Int. Conf. Mobile Comput. Netw.*, 2024, pp. 528–542.
[67] Y. Chen, J. Yu, L. Kong, Y. Zhu, and F. Tang, "Sensing human gait for environment-independent user authentication using commodity RFID devices," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 6304–6317, May 2024.
[68] T. Zheng, Z. Chen, C. Cai, J. Luo, and X. Zhang, "V2iFi: In-vehicle vital sign monitoring via compact RF sensing," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 4, no. 2, pp. 1–27, 2020.
[69] T. Zheng, Z. Chen, S. Zhang, C. Cai, and J. Luo, "MoRe-Fi: Motion-robust and fine-grained respiration monitoring via deep-learning UWB radar," in *Proc. 19th ACM Conf. Embedded Netw. Sensor Syst.*, 2021, pp. 111–124.
[70] Z. Yang, Y. Zhang, K. Qian, and C. S. Wu, "A spectrogram learning neural network for deep wireless sensing," in *Proc. 20th USENIX Symp. Netw. Syst. Des. Implementation*, Boston, MA, USA, 2023, pp. 17–19.
[71] K. Liang, A. Zhou, Z. Zhang, H. Zhou, H. Ma, and C. Wu, "mmStress: Distilling human stress from daily activities via contact-less millimeter-wave sensing," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 7, no. 3, pp. 1–36, 2023.
[72] Y. Tian, G.-H. Lee, H. He, C.-Y. Hsu, and D. Katabi, "RF-based fall monitoring using convolutional neural networks," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 2, no. 3, pp. 1–24, 2018.
[73] D. Zhang et al., "LT-Fall: The design and implementation of a life-threatening fall detection and alarming system," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 7, no. 1, pp. 1–24, 2023.
[74] Y. Li et al., "Towards domain-independent and real-time gesture recognition using mmWave signal," *IEEE Trans. Mobile Comput.*, vol. 22, no. 12, pp. 7355–7369, Dec. 2023.
[75] H. Liu et al., "Real-time arm gesture recognition in smart home scenarios via millimeter wave sensing," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 4, no. 4, pp. 1–28, 2020.
[76] J. Chen, D. Zhang, Z. Wu, F. Zhou, Q. Sun, and Y. Chen, "Contactless electrocardiogram monitoring with millimeter wave radar," *IEEE Trans. Mobile Comput.*, vol. 23, no. 1, pp. 270–285, Jan. 2024.
[77] F. Wang, X. Zeng, C. Wu, B. Wang, and K. R. Liu, "mmHRV: Contactless heart rate variability monitoring using millimeter-wave radio," *IEEE Internet Things J.*, vol. 8, no. 22, pp. 16623–16636, Nov. 2021.
[78] C. Xu et al., "CardiacWave: A mmWave-based scheme of non-contact and high-definition heart activity computing," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 5, no. 3, pp. 1–26, 2021.
[79] U. Ha, S. Madani, and F. Adib, "WiStress: Contactless stress monitoring using wireless signals," *Proc. ACM Interactive Mobile Wearable Ubiquitous Technol.*, vol. 5, no. 3, pp. 1–37, 2021.
[80] H. Yin et al., "Let IoT know you better: User identification and emotion recognition through millimeter-wave sensing," *IEEE Internet Things J.*, vol. 10, no. 2, pp. 1149–1161, Jan. 2023.
[81] J. Kim, G. Bae, C. Kim, J. Lee, W. Shin, and H. Oh, "Doppler correspondence: Non-iterative scan matching with Doppler velocity-based correspondence," 2025, *arXiv:2502.11461*.
[82] A. Segal, D. Haehnel, and S. Thrun, "Generalized-ICP," in *Proc. Robot.: Sci. Syst.*, 2009, vol. 2, no. 4, pp. 435–435.
[83] D. C. Herraez, M. Zeller, L. Chang, I. Vizzo, M. Heidingsfeld, and C. Stachniss, "Radar-only odometry and mapping for autonomous vehicles," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2024, pp. 10275–10282.
[84] Y. Xu, Q. Huang, S. Shen, and H. Yin, "Incorporating point uncertainty in radar SLAM," *IEEE Robot. Automat. Lett.*, vol. 10, no. 3, pp. 2168–2175, Mar. 2025.



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